RAHMAN WIDYANTO

Cikarang, Indonesia | +62 877-2335-7690| rahmanwidyanto01@gmail.com | www.linkedin.com/in/rahman-widyanto

SUMMARY

Mechatronics Engineer with 4+ years of hands-on experience in mobile robot technologies, specializing in preparation, integration, and site implementation of diverse robotic solutions. Skilled in managing various robot types and third-party peripherals, ensuring seamless interoperability and project success. Adept at both technical execution and managerial oversight, with a proven ability to troubleshoot unexpected challenges and deliver solutions that meet or exceed customer expectations. Recognized for versatility, problem-solving, and commitment to high-quality project delivery in dynamic environments.

WORK EXPERIENCES

PT. Artifa Sukses Persada (Infinitigroup) - Jakarta — Jan 2026 - Present

Scrum, Quality and Documents

System Integrator

Area of Responsibility: Conducted pre-delivery quality inspections of mechanical assemblies, electrical systems, and robotic equipment to ensure compliance with project requirements and delivery standards. Coordinated Factory Acceptance Test (FAT) activities, including test preparation, issue documentation, defect follow-up, and cross-functional communication with engineering and project teams, while providing hands-on technical support during troubleshooting to identify root causes and implement corrective actions prior to deployment.

Apr 2024 – Dec 2025

Principle and Product Engineer

System Integrator

Area of Responsibility: Delivered technical concept development for robotics and automation projects, covering requirements analysis, electrical diagram design, hardware communication architecture, system integration planning, robot programming, troubleshooting, and vendor coordination.

Aug 2022 – Mar 2024

Field Application Engineer

System Integrator

Area of Responsibility: Managed field deployment and commissioning of robotics and automation systems, ensuring equipment was properly installed, configured, tested, validated, and handed over in accordance with customer requirements, project scope, and agreed timelines.

Projects

HIKROBOT CTU | Indonesia | 2025

YKK Zipper Cimanggis — CTU Warehouse Automation

- Implemented site layouts, automation flows, and operational tuning based on on-site findings and system requirements.
- Troubleshot material-flow and robot-jamming issues and designed corrective solutions to improve process reliability.
- Optimized warehouse layout to reduce space consumption and simplify operational data management.
- Designed and implemented a custom safety gate to control robot movement in operator-access areas.
- Developed Node-RED and PLC communication with HIKROBOT RCS for safety interlocking and controlled robot operation.

Technologies: HIKROBOT CTU | RCS | Node-RED | PLC | Warehouse Automation

HIKROBOT AMR | Indonesia | 2024

Wilmar Medan — AMR Warehouse Automation

- Planned and implemented AMR layouts based on fleet size, site constraints, and throughput requirements.
- Assisted system integrators with AMR RCS integration and operational interface requirements.
- Recommended alternative materials and solutions to improve the durability of navigation QR markers.
- Designed a specialized QR layout and installation method to enable reliable AMR navigation into container boxes.
- Supported site optimization and validation of AMR navigation performance.

Technologies: HIKROBOT AMR | RCS | QR Navigation | Warehouse Automation

HIKROBOT AMR & FMR | Indonesia | 2023

Astra Honda Motor Cikarang — Multi-Robot AMR & FMR Integration

- Designed and implemented AMR navigation layouts and established standards for wide-area QR installations.
- Implemented 2D SLAM for AMR and 3D SLAM for FMR within the same operational area.
- Integrated AMR and FMR operations where both robot types shared and intersected operational zones.
- Developed PLC and Node-RED communication with HIKROBOT RCS and third-party software for robot interlocking.
- Implemented interlocking logic between the FMR, robotic arm, and supporting systems to ensure safe and reliable material transfer.

Technologies: HIKROBOT AMR | HIKROBOT FMR | 2D SLAM | 3D SLAM | RCS | Node-RED | PLC

HIKROBOT AMR & FMR | Indonesia | 2022

Shell Manufacturing Indonesia — AMR & FMR Automation

- Implemented wide-area QR navigation systems and established installation standards for large and intersecting QR matrices.
- Supported the initial implementation of AMR and FMR operating within the same environment.
- Developed Visual Studio applications to communicate with Neurala and Basler camera systems for automated inspection.
- Integrated PLC-controlled bridging conveyors with the robot system for automated material transfer.
- Contributed to establishing implementation standards for AMR navigation and multi-robot operations.

Technologies: HIKROBOT AMR | HIKROBOT FMR | QR Navigation | Visual Studio | Neurala | Basler Camera | PLC | Conveyor Integration

SEER FMR | Indonesia | 2024

EPSON Indonesia — FMR Warehouse Delivery Automation

- Implemented SEER FMR for automated warehouse deliveries, reducing manual material-handling processes and improving internal logistics efficiency.
- Oversaw network installation and configuration to establish reliable communication between the FMR and supporting systems.
- Configured and implemented SLAM-based navigation according to warehouse conditions and operational requirements.
- Designed and configured FMR process flows to support automated delivery operations and material movement.
- Conducted system testing, troubleshooting, and on-site tuning to ensure reliable FMR operation and process execution.

Technologies: SEER FMR | SLAM | Network Infrastructure | Warehouse Automation | Mobile Robotics

SKILLS

AGV and AMR Platforms: HIKROBOT, CASUN, LINDIN, SEER
Navigation & Mapping: QR Based Navigations, QR Matrix Design, 2D & 3D SLAM Mapping and Optimization, Navigations Troubleshooting
Programming Language: C#, Python, various robotics native software
Communication Protocols: MQTT, Modbus, REST API, Json, FINS, MC Protocol
PLC: Mitsubishi, OMRON, PUSR
Programming Tools: Visual Studio, Nodered
Languages: Bahasa Indonesia (native), English (proficient)

EDUCATION

Politeknik Manufaktur Bandung - Bandung — Aug 2016 - Jan 2022
Bachelor of Applied Science, 3.29 of 4.00

Major Courses: Control Systems, Feedback Control, Mechatronics, Robotics Fundamentals, Electric Motors and Actuators, Sensors and Instrumentation

Final Project: *Simulation of Multi-Robot Obstacle Avoidance with Fuzzy Logic Algorithm and YOLO Object Detection*